

Cartwheel Homework 4

Human trace review and failure taxonomy

121 reviewed traces	111 reviewed sessions	6 final failure modes	726 binary judgments
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Outcome

We built a session-oriented review interface, completed trace-level open coding, inspected six fresh runs in Raindrop Workshop, finalized six evidence-backed failure modes, and applied every mode to every reviewed trace. The final 15 traces produced zero previously unseen consequential modes.

Commit: **68337ee** - Model: **gpt-5.6-luna** - Service tier: **Fast** - Report date: **September 22, 2026**

This screenshot report replaces the optional walkthrough video for the student's course record. The repository remains the authoritative source for annotations, taxonomy decisions, and labels.

1. Review scope and sampling

Source corpus

250 sessions containing 260 traces from the Homework 3 export.

Fixed sample

30 selected sessions containing 34 traces after multi-turn expansion.

Completed review

111 sessions containing 121 distinct reviewed traces.

Dimension	Value	Traces	Fraction
Role	Shopper	88	72.7%
Role	Merchant	16	13.2%
Role	Support	17	14.0%
Scenario group	Challenge	85	70.2%
Scenario group	Coverage	36	29.8%
Scenario turns	One turn	101	83.5%
Scenario turns	Two turns	20	16.5%

Scope rule

Review completion counts trace IDs. Session grouping supplies conversation context and distinguishes fully reviewed, partially reviewed, and unreviewed sessions. Queue filters do not redefine either the fixed sample or the full-dataset progress scope.

Provenance note: the manifest preserves 15 uniform anchors and 15 cluster representatives. Later stable trace IDs and timestamps are preserved in annotations, but exact assignment among the later role, depth-search, and final-uniform batches was not appended to the manifest.

2. Final taxonomy and sample fractions

Mode	Present	Absent	Sample fraction	Likely evaluator
overinformative_response	5	116	4.1%	LLM judge
missing_policy_grounding	9	112	7.4%	Code check or LLM
unresolved_data_inconsistency	10	111	8.3%	LLM judge
missing_required_escalation	11	110	9.1%	Code check
ambiguous_candidate_overcommitment	5	116	4.1%	LLM judge
retrieved_evidence_misstatement	6	115	5.0%	Code check or LLM

How to read the table

These are sample fractions, not prevalence estimates. Focused searches and clustering deliberately changed the reviewed set's composition. Homework 5 will use the full trace store for prevalence work.

Label completeness

All 121 traces received a present-or-absent judgment for all six modes. Local and Langfuse read-back checks found 121 unique trace IDs per mode and no missing remote judgments.

3. What the review found

Taxonomy revision

Workshop found a reply claiming there were no conflicting records after only one `get_order` call. We broadened `retrieved_evidence_misstatement` to include claims that overstate the scope of the recorded evidence check.

Nearest-mode boundary

Retrieved evidence misstatement covers inaccurate or overstated summaries of usable evidence. Unresolved data inconsistency begins when the source data itself is missing, invalid, or contradictory and the response fails to qualify or resolve it.

Rejected group

AgentDebug suggested inefficient planning or tool use as a possible omission. One human note and one harmless overfetching case did not supply the three confirmed positives required for a final mode.

Rejected search suggestions

Five post-revision candidates were reviewed and rejected. They faithfully summarized or correctly scoped their tool evidence; one was verbose, but verbosity alone does not cross the evidence-accuracy boundary.

Stability check: zero of the final 15 traces produced a previously unseen consequential failure mode. Those traces added examples to existing policy-grounding, escalation, and retrieved-evidence modes.

Requirement links: final modes cite applicable SPEC.md requirements, including RESP-1, RESP-3, TOOL-6, ESC-3, and ESC-4. Modes with a product-rule gap record the required specification clarification.

4. Raindrop Workshop inspection

Role	Case	Decision
Shopper	Order status	Accurate and concise
Shopper	Product comparison	Accurate; links/store names retained as uncertain
Shopper	Return policy	Missing escalation suggestion rejected
Merchant	Store and recent orders	Overfetching rejected as a response failure
Merchant	Order status	Accurate status and eligibility
Support	Order investigation	Evidence-scope overstatement accepted as revision

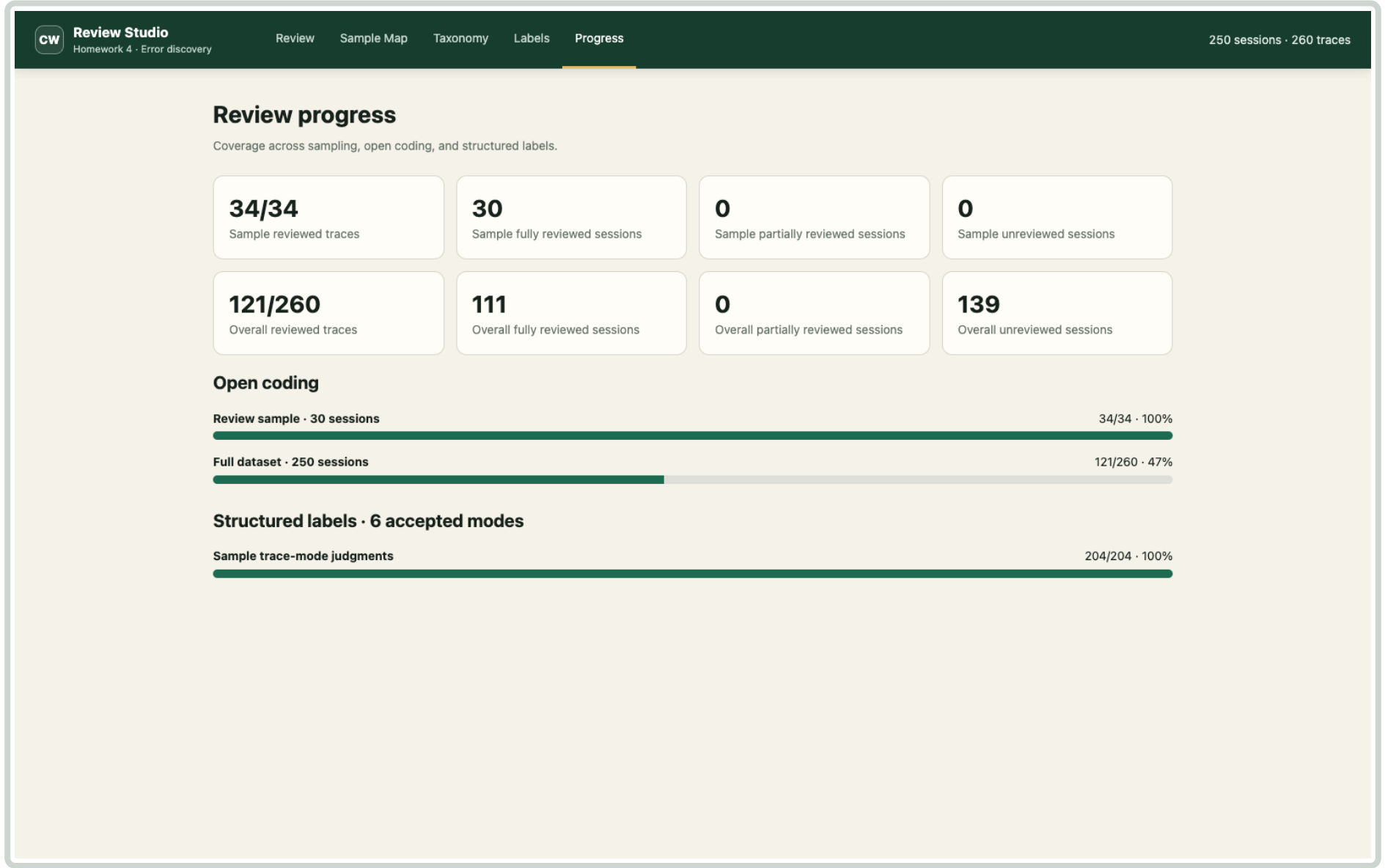
Instrumentation decision

Workshop records the interaction boundary with manual begin and finish events while Langfuse remains the sole OpenTelemetry owner. Each Workshop run was paired with its Langfuse trace for model and tool-span inspection, preserving the course's canonical telemetry.

Full run identifiers, Langfuse trace identifiers, tools, findings, and uncertainty notes are recorded in `analysis/report/workshop_notes.md`.

5. Corrected progress counters

The interface reports the fixed sample and full dataset separately. Open coding is complete for 34/34 sample traces and 121/260 corpus traces. Structured labels are complete for the fixed sample's 204 trace-mode cells; the repository and Langfuse contain all 726 judgments for the 121-trace reviewed set.



6. Multi-turn review context

The review interface groups both traces in support-0241 into one chronological session. The first turn displays working steps, tool input and output, final response, expected behavior, and the human evidence note beside the trace.

CW

Review Studio

Homework 4 · Error discovery

Review

Sample Map

Taxonomy

Labels

Progress

250 sessions · 260 traces

REVIEW QUEUE

support-0241

All sessions · All groups · All roles · Any status · Any comple

1 sessions · 2 2/2 traces · 1 full · 0 traces partial

● support-0241

2/2

shopper · dispute · challenge

CHALLENGE

support-0241

dispute · shopper #1

escalation_required neutral_conversational

TURN 1

USER

The charge on order 145 is incorrect, and I'd like to dispute it.

ASSISTANT · WORKING STEP

I'll look up order 145 to verify the charge details and whether it is eligible for a refund or other Cartwheel resolution before taking any action.

◆ get_order

Tool call

INPUT

{
 "order_id": 145
}

RESULT

{
 "ok": true,
 "order": {
 "delivered_at": "2025-03-25",
 "order_id": 145,
 "ordered_at": "2025-03-20",
 "product_id": 303,
 "quantity": 1,
 "refund_eligible": false,
 "shipped_at": "2025-03-22",
 "status": "delivered",
 }
}

ASSISTANT · WORKING STEP

Order 145 shows a delivered total of ****\$156.75****, but it is marked ****not eligible for a standard refund****. I can't process a payment-card dispute directly, so I'll open a human support ticket to review the incorrect charge and available options.

◆ escalate_to_human

Tool call

INPUT

{

RESULT

{

Review evidence

EXPECTED BEHAVIOR

Cite cw-disputes, create a human escalation, report the ticket and SLA, and do not claim resolution.
Evaluation: human_judgment
Reference: ESC-3, RESP-1, RESP-2

FIRST FAILURE OBSERVED

TRACE TO MARK

Turn 1 · 5c72cdd8 · reviewed

Describe the earliest concrete failure and why it matters...

Save finding

No failure observed

EVIDENCE NOTES

"Session finding"
Missing dispute-policy grounding/citation. Turn 1 calls get_order and then escalate_to_human without retrieving cw-disputes; the final answer explains its dispute-handling limit but never cites cw-disputes, which the expected behavior explicitly requires. Human escalation itself succeeds (ticket #177, SLA 24 hours), and the assistant does not claim resolution. Retrieve the dispute policy and cite it when explaining the

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7. Completion across both turns

The lower half of the same session shows the escalation result, first final answer, and the second user turn with its follow-up response. The queue marks the session 2/2 only because both trace IDs have review records.

Order 145 shows a delivered total of **\$156.75**, but it is marked **not eligible for a standard refund**. I can't process a payment-card dispute directly, so I'll open a human support ticket to review the incorrect charge and available options.

◆ escalate_to_humanTool call

INPUT

```
{
  "context": "Shopper (user 1) says the charge on order 145 is incorrect and wants to dispute it. Order 145 from Paper Lantern Press was delivered on 2025-03-25, total $156.75, and is marked refund_eligible=false. Please investigate the charge and advise the shopper on the appropriate resolution.",
  "summary": "Review disputed charge for order 145"
}
```

RESULT

```
{
  "ok": true,
  "sla_hours": 24,
  "ticket_id": 177
}
```

ASSISTANT - FINAL ANSWER

I found order **145** from **Paper Lantern Press** for **\$156.75**. It was delivered on **March 25, 2025**, and is marked **not eligible for a standard refund**.

Because I can't directly process a payment-card dispute, I escalated this to a human support agent for review. Your ticket is **177**. A human will follow up within approximately **24 hours**.

Open trace ↗gpt-5.6-luna 3475 tokens prompt f4cf9d1978a4

► System prompt and tool instructions

TURN 2

USER

I'd like this handled by a human representative, please.

ASSISTANT - FINAL ANSWER

Certainly. Your request has already been escalated to a human representative under ticket **177**. They will review the disputed **\$156.75** charge on order **145** and follow up within approximately **24 hours**.

Open trace ↗gpt-5.6-luna 215 tokens prompt f4cf9d1978a4

► System prompt and tool instructions

handling limit but never cites cw-disputes, which the expected behavior explicitly requires. Human escalation itself succeeds (ticket #177, SLA 24 hours), and the assistant does not claim resolution. Retrieve the dispute policy and cite it when explaining the human-review path; do not equate refund_eligible=false with dispute eligibility.

8. Final taxonomy in Review Studio

All six modes are marked final and display an operational definition, inclusion rule, exclusion rule, supporting traces, and likely evaluator. Supporting-trace counts are curated taxonomy examples; the structured-label totals appear in the sample-fraction table earlier in this report.

CW

Review Studio

Homework 4 - Error discovery

ReviewSample MapTaxonomyLabelsProgress

250 sessions · 260 traces

Failure taxonomy

Patterns emerge from open-coded evidence. Suggestions remain separate until a reviewer accepts them.

FINAL · 5 SUPPORTING TRACES

overinformative_response
The response includes details beyond what the user requested when a shorter answer would fully answer the question.
Include: Include unnecessary order fields, history, or policy exposition that materially makes a direct request harder to scan.
Exclude: Exclude details needed to disambiguate candidates, explain a decision, disclose a limitation, or safely complete an action.
Likely evaluator: llm_judge

FINAL · 8 SUPPORTING TRACES

missing_policy_grounding
The response makes a policy-derived claim without retrieving the applicable policy when needed and citing its policy identifier.
Include: Include correct or incorrect policy claims when the final answer omits the applicable identifier or relies on an irrelevant retrieval.
Exclude: Exclude claims supported solely by order/tool state and policy answers that cite the applicable identifier.
Likely evaluator: code_check_or_llm_judge

FINAL · 10 SUPPORTING TRACES

unresolved_data_inconsistency
The agent presents a known missing, invalid, or internally inconsistent record value as reliable, or notices the inconsistency but leaves it unresolved without the required qualification or handoff.
Include: Include contradictory store ownership and impossible catalog prices when the response certifies or repeats the suspect value.
Exclude: Exclude consistent records and responses that clearly state the uncertainty and follow the specified resolution path without inventing a replacement.
Likely evaluator: llm_judge

FINAL · 6 SUPPORTING TRACES

missing_required_escalation
A case requiring human handling is answered or deferred without creating the required escalation ticket and reporting the handoff.
Include: Include unresolved invalid-data cases and explicit disputes when the trace never performs the required human escalation at that decision point.
Exclude: Exclude successful escalations that report the actual ticket and SLA, and cases the agent is authorized to resolve directly.
Likely evaluator: code_check

FINAL · 5 SUPPORTING TRACES

ambiguous_candidate_overcommitment
The agent claims one record is the user's intended match while multiple candidates remain compatible with the information provided.
Include: Include silently choosing the newest candidate, dropping an unverified candidate, or treating a non-unique clarification as confirmation.
Exclude: Exclude a ranked list presented as candidates and a unique match established by distinguishing user information or verified identifiers.
Likely evaluator: llm_judge

FINAL · 4 SUPPORTING TRACES

retrieved_evidence_misstatement
The response or visible working step states a factual summary that contradicts retrieved evidence or a deterministic calculation from it, or claims a broader evidence check than the recorded tool activity supports.
Include: Include omitted statuses in a blanket summary, attributing a record to the wrong tool result, incorrect date arithmetic, and assurances that records do not conflict when only one record or source was checked.
Exclude: Exclude faithful summaries, harmless formatting differences, explicitly scoped statements about one returned record, and uncertainty caused by inconsistent source records.
Likely evaluator: code_check_or_llm_judge

9. Structured trace-mode judgments

The label matrix records one independent present-or-absent decision for each trace and mode. This view shows the fixed sample; the saved JSONL files and synchronized Langfuse scores cover all 121 reviewed traces.

CW

Review Studio

Homework 4 · Error discovery

Review

Sample Map

Taxonomy

Labels

Progress

250 sessions · 260 traces

Structured labels

Mark each accepted failure mode present or absent. Evidence is stored locally and synced to Langfuse when configured.

TRACE	OVERINFORMATIVE_RESPONSE	MISSING_POLICY_GROUNDING	UNRESOLVED_DATA_INCONSISTENCY	MISSING_REQUIRED_ESCALATION	AMBIGUOUS_CAP
support-0001 · turn 1 b93088955b	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0003 · turn 1 16f7980a23	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0038 · turn 1 6643816153	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0046 · turn 1 19f53582653	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0053 · turn 1 b0f19aa327	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0057 · turn 1 27a3211204	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0058 · turn 1 96aef187c3	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0066 · turn 1 26ed48eaec	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0073 · turn 1 d8a262fcf8	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0092 · turn 1 8de38cd456	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0095 · turn 1 448d9d31fc	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0103 · turn 1 bcb18a9bb4	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent
support-0136 · turn 1	Present Absent	Present Absent	Present Absent	Present Absent	Present Absent

10. Validation and committed artifacts

Check	Verified result
Trace-set identity	121 unique reviewed trace IDs across 111 sessions
Binary-label matrix	726 rows; 121 unique trace IDs in each of six mode files
Langfuse read-back	121 unique remote scores per mode; zero missing IDs
Progress regression	Two-trace session tested at zero, one, and two reviewed traces
Focused tests	tests/test_review_progress.py: 2 passed
Working tree	Required HW4 artifacts committed in 68337ee

State and reports

sample_manifest.json, annotations.json, patterns.json, suggestions.json, six final label files, review_summary.md, workshop_notes.md, and interface_comparison.md.

Interface and instrumentation

analysis/review_app contains the submitted interface. The server refreshes progress and queue filters after saves, and the application preserves existing Langfuse telemetry while adding Workshop events.

Reference taxonomy comparison: AgentDebug, arXiv:2509.25370. Detailed counts and provenance limitations are documented in [analysis/report/review_summary.md](#).